Leon Riccius | Curriculum Vitae

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Education

Doctor of Philosophy — **Delft University of Technology, NL**
Civil Engineering — *07/2021 – present*

Thesis: Efficient Bayesian Inference of Parametric Partial Differential Equations
Supervisors: Dr. Iuri Rocha & Dr. Frans P. van der Meer

Master of Science — **Technical University Munich, GER**
Mechanical Engineering — *03/2018 – 05/2021*

Thesis: Machine Learning Augmented Turbulence Modelling for the Reynolds Stress Closure Problem
Supervisor: Prof. P.S. Koutsourelakis

Term Paper: Advanced Timestepping Techniques for an Incompressible Navier-Stokes Solver with Discontinuous Galerkin Discretization
Supervisors: Prof. W.A. Wall & Prof. M. Kronbichler

Bachelor of Science — **Technical University Munich, GER**
Mechanical Engineering — *10/2013 – 02/2018*

Thesis: Diffeomorphic Matching on Thrombus and Outer Surfaces of Abdominal Aortic Aneurysms
Supervisor: Prof. M.W. Gee

Semesters abroad:
Keio University, Tokyo, JP — *03/2019 – 09/2019*
Ecole Polytechnique de Montréal, Montréal, CA — *01/2016 – 05/2016*

Publications

Articles

[1] Leon Riccius, Iuri Rocha, Joris Bierkens, Hanne Kekkonen, and Frans van der Meer. Integration of Active Learning and MCMC Sampling for Efficient Bayesian Calibration of Mechanical Properties. *European Journal of Mechanics A/Solids*, 2026.

[2] Leon Riccius, Iuri Rocha, Joris Bierkens, Hanne Kekkonen, and Frans van der Meer. Piecewise Deterministic Markov Processes for Bayesian Inference of PDE Coefficients. 2026. under review.

Conference Papers

[3] Leon Riccius, Atul Agrawal, and Phaedon-Stelios Koutsourelakis. Physics-Informed Tensor Basis Neural Network for Turbulence Closure Modeling. In *37th Conference on Neural Information Processing Systems, NeurIPS*, 2023.

Conference Talks

[4] Leon Riccius, Iuri Rocha, Joris Bierkens, Hanne Kekkonen, and Frans van der Meer. Zig-Zag Sampling for Bayesian Inference in Parametric PDEs. In *Third Joint SIAM/CAIMS Annual Meeting*, 2025.

[5] Leon Riccius, Iuri Rocha, Joris Bierkens, Hanne Kekkonen, and Frans van der Meer. Zig-Zag Sampling for Bayesian Inference of Material Parameter Fields. In *6th International Conference on Uncertainty Quantification in Computational Science and Engineering, UNCECOMP*, 2025.

[6] Leon Riccius, Iuri Rocha, and Frans van der Meer. Comparison of Surrogate Modeling and Sampling Strategies for Efficient Bayesian Model Calibration. In *5th International Conference on Uncertainty Quantification in Computational Science and Engineering, UNCECOMP*, 2023.

[7] Leon Riccius, Iuri Rocha, and Frans van der Meer. Efficient Bayesian Inference for Parametric pde Inverse Problems using Zig-Zag Sampling and Approximate Gradients. In *9th European Congress on Computational Methods in Applied Sciences and Engineering, ECCOMAS*, 2024.

[8] Leon Riccius, Iuri Rocha, and Frans van der Meer. Zig-Zag Sampling with Approximate Event Rates for Bayesian Inference of Material Parameters. In *1st Dutch Computational Science Day, DUCOMS*, 2024.

Invited Talks

[9] Leon Riccius. Comparing Surrogate Modeling and Sampling Strategies for Efficient Bayesian Model Calibration. In *Eurandom Workshop Bayesian Computation*. 2024.

Teaching

TU Delft, Faculty for Civil Engineering and Geoscience **Delft, NL**

Designing interactive textbook[1], teaching assistant[2], project supervisor[3] *07/2021 – present*

- Modeling, Uncertainty, and Data Analysis for Engineers[1,2,3]
- Data Science and Artificial Intelligence for Engineers[1,2,3]
- Statistical Learning for Engineers[2]

Supervision:

Daan Smolders, Master Thesis, in cooperation with Airbus 2024

Surrogate modelling for uncertainty propagation by targeting regions of high variance and probability

TU Munich, Department of Mathematics **Munich, GER**

Teaching assistant, bachelor level *08/2016 – 10/2021*

- Various analysis and algebra tutoring classes for Mechanical and Electrical Engineering
- Lecture exercises with up to 500 participants

Practical Experience

TU Munich, Department of Mechanical Engineering,
Chair of Metal Forming and Casting **Munich, GER**

Academic Assistant, 50% *04/2021 – 06/2021*

- 3D data processing in Python for reverse engineering in metal forming
- Sparse representations of laser scanner point clouds via uniform and non-uniform B-Splines

BMW Group **Munich, GER**

Student Intern, 100% *10/2015 – 12/2015*

- Ensuring of target agreement compliance within the project development of the 6*er*-series
- Endurance test coordination in the engine design department

Awards

Travel Award — **Montréal, CA**
Society for Industrial and Applied Mathematics (SIAM) — *2025*

Best Poster Award — **Papendal, NL**
27th Engineering Mechanics Symposium (114 submissions) — *2024*

Software skills

Programming:	Python (pytorch, jax), C++, C, bash, agentic coding
Modelling & Analysis:	MatLab, OpenFOAM, ANSYS
Software & Tools:	Git, LaTeX, Paraview
General:	Data visualization, academic writing, public speaking, supervision

Languages

German:	Native proficiency	
English:	Full professional proficiency	*TOEFL iBT Score: 111*
French:	Limited working proficiency	*Abitur: B2/C1*
Dutch:	Elementary proficiency	
Japanese:	Elementary proficiency	